



Modelling Financial Markets

Chapter 6: Multivariate Volatility and Dependence



Daniel Traian PELE

Bucharest University of Economic Studies

IDA Institute Digital Assets

Blockchain Research Center

AI4EFin Artificial Intelligence for Energy Finance

Romanian Academy, Institute for Economic Forecasting

MSCA Digital Finance

Acronyms used in this material (1/2)

- ▣ ACWI: All Country World Index (MSCI)
- ▣ AI: Artificial Intelligence
- ▣ AIC: Akaike Information Criterion
- ▣ AR: AutoRegressive (model)
- ▣ ARCH: AutoRegressive Conditional Heteroskedasticity
- ▣ BAC: Bank of America Corporation (NYSE ticker)
- ▣ BEKK: Baba–Engle–Kraft–Kroner (multivariate GARCH model)
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BIC: Bayesian Information Criterion
- ▣ BNP: BNP Paribas SA (Euronext Paris ticker; Banque Nationale de Paris)
- ▣ CCC: Constant Conditional Correlation (model)
- ▣ CDO: Collateralised Debt Obligation
- ▣ CML: Canonical Maximum Likelihood
- ▣ COVID-19: COronaVirus Disease 2019
- ▣ COVOL: COmmon VOLatility (global COVOL: the size of the volatility shock common to all assets)
- ▣ CPI: Consumer Price Index
- ▣ DBK: Deutsche Bank AG (Xetra ticker)
- ▣ DCC: Dynamic Conditional Correlation
- ▣ DCC-NL: Dynamic Conditional Correlation with NonLinear shrinkage
- ▣ EPU: Economic Policy Uncertainty (index)
- ▣ ES: Expected Shortfall
- ▣ ETF: Exchange-Traded Fund
- ▣ EU: European Union
- ▣ EWMA: Exponentially Weighted Moving Average
- ▣ FRM: Financial Risk Meter
- ▣ GARCH: Generalised AutoRegressive Conditional Heteroskedasticity

Acronyms used in this material (2/2)

- GAS: Generalised Autoregressive Score (model)
- GPR: GeoPolitical Risk (index)
- HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- IFM: Inference Functions for Margins
- JPM: JPMorgan Chase & Co. (NYSE ticker)
- LR: Likelihood Ratio (test)
- MSCI: Morgan Stanley Capital International (index provider)
- NYSE: New York Stock Exchange
- PC1: first Principal Component
- SE: Standard Error
- US: United States
- VEC: VECtorised multivariate GARCH model (vech of the covariance matrix)
- VIX: CBOE Volatility Index

Chapter Route and Outcomes

Route of this chapter (2×80 minutes)

- ▣ Session 1: why dependence matters, covariance estimation and its uncertainty, multivariate GARCH and DCC, crisis correlations and the Forbes–Rigobon bias
- ▣ Session 2: copulas and tail dependence, applications to Europe, Romania and banks, a case study on global common volatility, one pass of the AI-assisted research loop
- ▣ Builds on Chapter 1 (fat tails, volatility clustering), Chapter 4 (Σ in portfolio choice) and Chapter 5 (univariate GARCH); leads to Chapter 7 (portfolio VaR and ES), Chapter 8 (backtesting) and Chapter 18 (systemic risk)
- ▣ Technical detail sits in the Appendix at the end

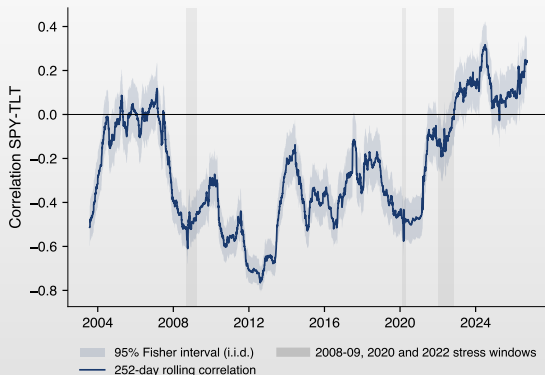
After this chapter you can

- ▣ estimate a time-varying covariance (rolling window, EWMA, DCC) and attach a valid standard error to a correlation
- ▣ separate contagion from the volatility bias of crisis correlations
- ▣ choose, fit and test a copula, and read its tail dependence at a finite threshold and in the limit

Why Dependence Matters: Portfolios, Hedges, Contagion

- **Diversification** (Markowitz (1952)): portfolio variance $\sigma_p^2 = w^\top \Sigma w$; with two assets the correlation enters only the cross term $2w(1-w)\rho\sigma_1\sigma_2$
 - ▶ SPY (S&P 500 exchange-traded fund) and TLT (20+ year US Treasury bond fund), 60/40: volatility 10.4% with the 2002–2021 correlation (-0.40), 13.2% with the 2022–2026 one (0.11), same asset volatilities
- **Hedging**: the minimum-variance hedge ratio $h^* = \rho\sigma_S/\sigma_F$ is the slope of r_S on r_F ; its effectiveness is ρ^2
 - ▶ If the correlation moves, yesterday's hedge is wrong today: a dynamic hedge needs a dynamic ρ_t
- **Contagion** (Forbes & Rigobon (2002)): a *significant increase* in cross-market linkages after a shock; **interdependence**: links present in all states (trade, common investors, common macro factors)
 - ▶ Correlations tend to rise exactly when diversification is needed most
- New in this chapter: Σ_t moves; how to estimate it, and how to measure the sampling uncertainty of every dependence estimate

Stocks and Bonds: A Correlation That Changed Sign



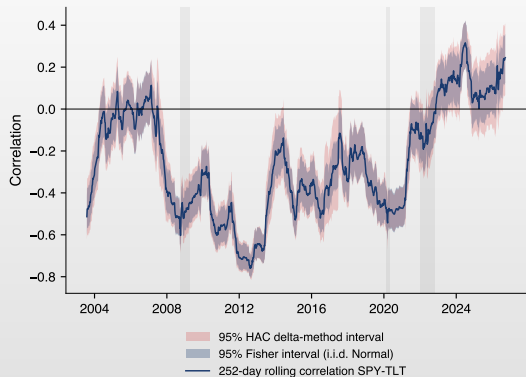
- Daily SPY and TLT returns, 252-day window; band: 95% Fisher interval assuming i.i.d. data; shaded: autumn 2008 – March 2009, February–April 2020, January–October 2022
- Minimum -0.77 (8 August 2012), maximum 0.32 (1 July 2024), last 0.24
- Positive on 6% of days before 2022, on 81% since January 2022
- Growth shocks move stocks and bond prices in opposite directions (negative correlation); inflation shocks hit both (positive, as in 2022)
- The Fisher band is too narrow here: it ignores fat tails and volatility clustering (next section)

 MFM_ch6_spy_tlt_rolling

Sample Covariance and Inference on a Correlation

- ▣ Returns $r_t \in \mathbb{R}^N$, $t = 1, \dots, T$: $\hat{\Sigma} = \frac{1}{T-1} \sum_t (r_t - \bar{r})(r_t - \bar{r})^\top$, Pearson $\hat{\rho}_{ij} = \hat{\Sigma}_{ij} / \sqrt{\hat{\Sigma}_{ii} \hat{\Sigma}_{jj}}$; $N(N+1)/2$ distinct elements, 125 250 for $N = 500$
- ▣ Fisher: $z = \text{artanh } \hat{\rho} \approx N(\text{artanh } \rho, 1/(T-3))$ for i.i.d. pairs with the bivariate Normal distribution
 - ▶ 95% interval $\tanh(z \pm 1.96/\sqrt{T-3})$; $T = 252$, $\rho = 0.5$: half-width about 0.09
- ▣ Under volatility clustering, the delta method: $\hat{\rho} - \rho \approx \frac{1}{T} \sum_t \psi_t$, $\psi_t = \tilde{x}_t \tilde{y}_t - \frac{\rho}{2}(\tilde{x}_t^2 + \tilde{y}_t^2)$, $\tilde{x}, \tilde{y}$ standardised
 - ▶ se from the long-run variance of ψ_t (HAC, Bartlett kernel, L lags, [Newey & West \(1987\)](#)); needs stationarity, finite fourth moments and weak dependence
 - ▶ i.i.d. Normal pairs give $\text{Var}(\psi_t) = (1 - \rho^2)^2$, the Fisher variance; fat tails and GARCH raise it (derivation in the Appendix)
- ▣ SPY-TLT daily, 2002–2021 ($T = 4\,891$, $\hat{\rho} = -0.403$): se 0.0120 (Fisher), 0.0217 (fourth moments, $L = 0$), 0.0227 (HAC, $L = 9$)

Fisher versus HAC Intervals: SPY and TLT



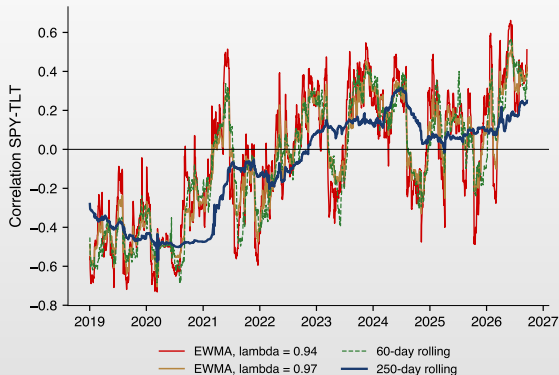
- 252-day rolling correlation, every 5th day, with the 95% Fisher interval and the 95% HAC delta-method interval (Bartlett, $L = 4$)
- HAC is wider in almost every window: median width ratio 1.24, 90% quantile 1.54
- Correlation significantly different from zero: 73% of windows by Fisher, 65% by HAC; the verdict flips in 8.2%
- Whole periods: HAC $se = 1.9 \times$ Fisher in 2002–2021, $1.4 \times$ since 2022; fourth moments alone give $1.8 \times$: fat tails matter more than autocorrelation
- Rule for daily returns: report delta-method HAC intervals

Rolling Windows and EWMA

- **Rolling window** of length m : $\hat{\Sigma}_t = \frac{1}{m} \sum_{s=t-m}^{t-1} r_s r_s^\top$ (mean set to zero for daily data)
 - ▶ Positive semidefinite, since $v^\top \hat{\Sigma}_t v = \frac{1}{m} \sum_s (v^\top r_s)^2 \geq 0$; singular if $m < N$, positive definite only if the m return vectors span $\mathbb{R}^N$
 - ▶ **Ghost effect**: an extreme day enters with weight $1/m$ and leaves abruptly m days later
- **EWMA** (exponentially weighted moving average, [J.P. Morgan/Reuters \(1996\)](#)):

$$\Sigma_t = \lambda \Sigma_{t-1} + (1 - \lambda) r_{t-1} r_{t-1}^\top = (1 - \lambda) \sum_{k \geq 1} \lambda^{k-1} r_{t-k} r_{t-k}^\top$$
 - ▶ $\lambda = 0.94$ for daily and 0.97 for monthly data; half-life $\ln 0.5 / \ln \lambda$: 11.2 and 22.8 periods; effective sample $(1 + \lambda)/(1 - \lambda) \approx 32$ days for $\lambda = 0.94$
 - ▶ Positive semidefinite, no ghost effect; but one λ for all elements and no mean reversion: an integrated GARCH(1,1) without a constant (Chapter 5)
- One trade-off for both: a short window or a small λ reacts fast but is noisy; a long window or a large λ is stable but slow to detect a regime change

EWMA versus Rolling Windows, SPY-TLT since 2019

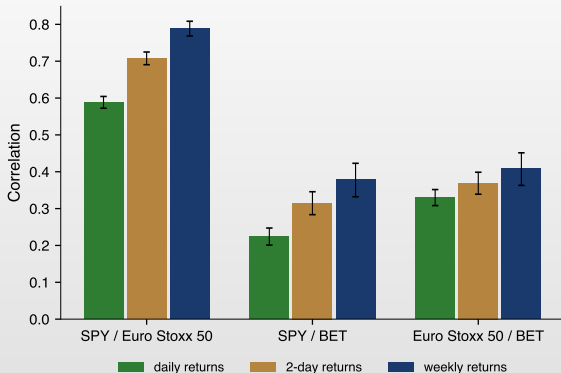


- Two EWMA estimators ($\lambda = 0.94$ and 0.97) and two rolling windows (60 and 250 days) on the same data
- Standard deviation of the daily change of the estimate: 0.060 (EWMA 0.94), 0.032 (EWMA 0.97), 0.0088 (250-day window)
- Positive for 20 days in a row from 10 May 2022 with EWMA 0.94, but only from 8 December 2022 with the 250-day window
- λ and m are fixed in advance; DCC (next section) estimates its own speed and memory (a, b)

Asynchronous Trading: A Trap for Daily Correlations

- Closing times differ: Bucharest and Frankfurt close hours before New York
 - ▶ US news after the European close enters European prices only the next day
- Consequence (positive dependence, as between equity markets): the same-day correlation is biased towards zero, and the lagged correlation is positive
- Remedies
 - ▶ Lower frequency: 2-day or weekly returns on common trading days
 - ▶ Add the lead-lag term: $\rho_0 + \rho_{+1}$, a rough approximation of the synchronous correlation, valid only for a one-day lag, no autocorrelation and equal variances
- Rule used throughout: join the *prices* on common days first, then compute returns

Daily, 2-Day and Weekly Correlations, 2000–2026



- SPY, Euro Stoxx 50 and BET on common trading days; bars: 95% Fisher intervals
- SPY–Euro Stoxx 50: 0.59 daily, 0.71 on 2-day returns, 0.79 weekly; SPY yesterday with Euro Stoxx 50 today: 0.20
- SPY–BET: 0.22 daily versus 0.38 weekly (lagged 0.17); Euro Stoxx 50–BET, almost synchronous closes: 0.33 versus 0.41
- Part of the “low” US–Europe daily correlation is a measurement artefact
- Weekly data are cleaner but give 1 367 instead of 6 446 observations: wider intervals

From Univariate to Multivariate Volatility

- Chapter 5: $r_t = \sigma_t z_t$, $\sigma_t^2 = \omega + \alpha r_{t-1}^2 + \beta \sigma_{t-1}^2$; now $r_t = H_t^{1/2} z_t$ with H_t known at $t-1$, $\mathbb{E}(z_t | \mathcal{F}_{t-1}) = 0$, $\text{Var}(z_t | \mathcal{F}_{t-1}) = I_N$, so $\text{Var}(r_t | \mathcal{F}_{t-1}) = H_t$
- Any model for H_t must deliver
 - ▶ positive definiteness at every t ; few parameters; interpretable dynamics
- Gaussian quasi-log-likelihood:

$$\ell = -\frac{1}{2} \sum_t (N \ln 2\pi + \ln |H_t| + r_t^\top H_t^{-1} r_t)$$
- **VEC** (Bollerslev, Engle & Wooldridge (1988)): every element of H_t on all squares and cross-products, positive definiteness not guaranteed; **BEKK** (Engle & Kroner (1995)): positive definite by construction, hard to interpret (recursions in the Appendix)

Number of parameters

N	VEC	full BEKK	DCC
2	21	11	$6 + 2$
5	465	65	$15 + 2$
10	6 105	255	$30 + 2$
50	3 252 525	6 275	$150 + 2$

- DCC: $3N$ univariate GARCH parameters plus 2, with the correlation target estimated separately

Constant and Dynamic Conditional Correlation

- Decomposition: $H_t = D_t R_t D_t$, $D_t = \text{diag}(\sigma_{1t}, \dots, \sigma_{Nt})$ from univariate GARCH; standardised residuals $\varepsilon_{it} = r_{it}/\sigma_{it}$
- CCC (Bollerslev (1990)): $R_t = R$ constant
- DCC (Engle (2002)): a GARCH-type recursion for a proxy matrix Q_t
 - ▶ $Q_t = (1 - a - b) \bar{Q} + a \varepsilon_{t-1} \varepsilon_{t-1}^\top + b Q_{t-1}$, $a, b \geq 0$, $a + b < 1$
 - ▶ $R_t = \text{diag}(Q_t)^{-1/2} Q_t \text{diag}(Q_t)^{-1/2}$
 - ▶ $\bar{Q}$: sample covariance of ε_t (correlation targeting)
- Two parameters govern all $N(N-1)/2$ correlations: parsimonious, but every pair has the same persistence
- Q_t is a covariance proxy, not a correlation: $\mathbb{E}_{t-1}[\varepsilon_t \varepsilon_t^\top] = R_t \neq Q_t$

Two-Step Estimation and Inference

- The Gaussian log-likelihood splits into a volatility part and a correlation part

$$\blacktriangleright \ell = \underbrace{-\frac{1}{2} \sum_t (N \ln 2\pi + 2 \ln |D_t| + r_t^\top D_t^{-2} r_t)}_{\ell_V} - \underbrace{\frac{1}{2} \sum_t (\ln |R_t| + \varepsilon_t^\top R_t^{-1} \varepsilon_t - \varepsilon_t^\top \varepsilon_t)}_{\ell_C}$$

- Step 1: fit N univariate GARCH(1,1) models, keep σ_{it} and ε_{it} ; step 2: fix $\bar{Q}$ at the sample covariance of ε_t and maximise ℓ_C over (a, b)
- Quasi-likelihood: Normality of z_t is not needed if conditional means and covariances are correct; but sample-covariance targeting makes the standard DCC estimator inconsistent in general, the corrected DCC is not (Aielli (2013))
- Step-2 standard errors treat the GARCH estimates and $\bar{Q}$ as known: propagate their error with the stacked sandwich of Newey & McFadden (1994) or by bootstrapping both steps (Appendix)
- CCC against DCC: $H_0 : a = 0$; b is not identified under H_0 , so the χ^2 reference is only indicative: bootstrap under CCC

DCC Estimates for SPY and TLT

Step 1: GARCH(1,1), returns in %

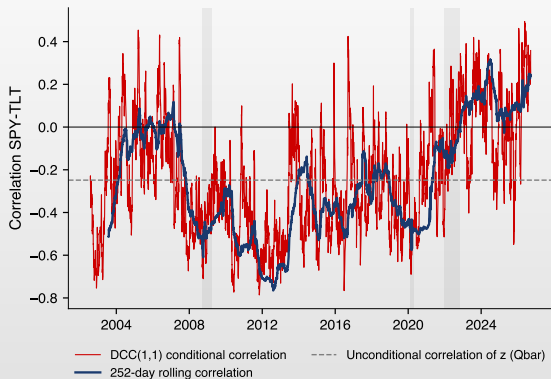
	ω	α	β	$\alpha + \beta$
SPY	0.0283	0.127	0.848	0.975
TLT	0.0057	0.049	0.943	0.993

Step 2: DCC(1,1) and three standard errors

	estimate	Hessian	boot. step 2	boot. both
a	0.0522	0.0059	0.0051	0.0052
b	0.9335	0.0084	0.0077	0.0077
$a + b$	0.9857	—	0.0038	0.0040

- Daily data 31 July 2002 – 18 September 2026, $T = 6073$; target $\bar{Q}_{12} = -0.248$
- Persistence $a + b = 0.986$: a shock to Q_t halves in ≈ 48 days, approximately, since $\mathbb{E}_{t-1}[\varepsilon_t \varepsilon_t^\top] = R_t \neq Q_t$
- LR against CCC 533.4; bootstrap of both steps under CCC ($B = 199$): largest null value 15.7, $p = 0.005$: CCC rejected
- Step 1 raises the se by only 2% (a) and 1% (b): T is large; the Hessian $\text{se}(\hat{b})$ is conservative here
- TLT volatility is more persistent (larger β) but reacts less to news (smaller α) than SPY volatility

DCC Correlation of SPY and TLT, 2002–2026

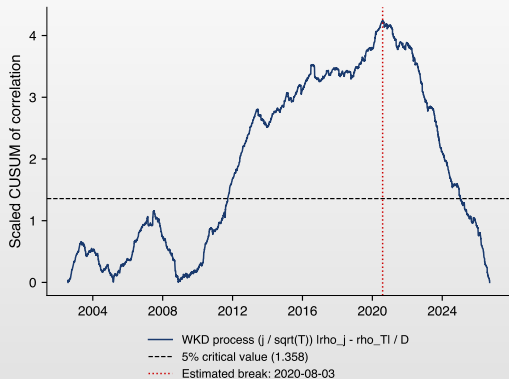


- DCC(1,1) conditional correlation, the 252-day rolling correlation and the target $\bar{Q}_{12}$
- Range -0.79 (2 November 2011) to 0.49 (21 May 2026); last value 0.36
- Average: -0.54 from 15 September to 31 December 2008, -0.56 in 2012, 0.00 in 2022
- DCC reacts within days, like EWMA, but its speed a and memory b are estimated
- Since January 2022 the path is above the target on 92% of days (36% before): a single long-run correlation is doubtful

Testing Constant Correlation at an Unknown Date

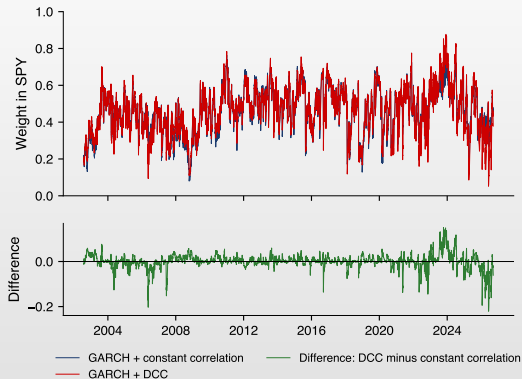
- Tse (2000): a Lagrange multiplier test of constant conditional correlation, computed from the CCC fit alone
- Picking a break date from a chart and then testing it with a fixed-date Fisher z test is oversized: the date is itself a function of the data
- Wied, Krämer & Dehling (2012): $Q_T = \max_{2 \leq j \leq T} \frac{j}{\sqrt{T} \hat{D}} |\hat{\rho}_j - \hat{\rho}_T|$, $\hat{\rho}_j$ the correlation of the first j observations
 - ▶ $\hat{D}^2$: the delta-method long-run variance of $\hat{\rho}$, Bartlett kernel with bandwidth $\lfloor T^{1/4} \rfloor$
 - ▶ Under H_0 (constant correlation, means and variances): $Q_T \rightarrow \sup_{0 \leq s \leq 1} |B(s)|$, B a Brownian bridge; 5% critical value 1.358
 - ▶ Constant variances are required, so we apply it to GARCH-standardised residuals; break date estimate $\hat{j} = \arg \max$

Dating the SPY-TLT Correlation Break



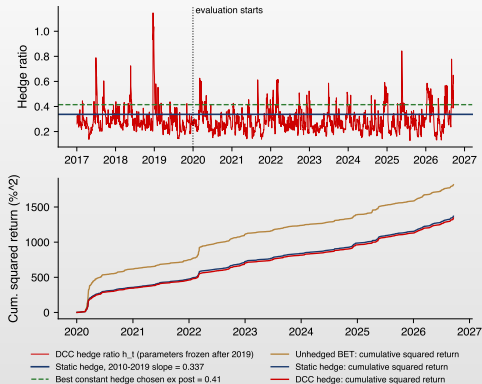
- Wied-Krämer-Dehling process on the GARCH(1,1)-standardised residuals, $T = 6\,073$, bandwidth 8; dashed: 5% critical value
- $Q_T = 4.25$ against 1.358: constant correlation rejected ($p < 0.001$)
- Estimated break 3 August 2020, not January 2022: residual correlation -0.36 before, 0.07 after
- Binary segmentation: a second break on 9 July 2007 ($Q = 3.01$); after the main break $Q = 1.56$, $p = 0.015$
- The limit law is proved for observed series: on estimated residuals the p -values are approximate; one DCC target over 2002–2026 averages at least two regimes

Application: Minimum-Variance Weights with DCC



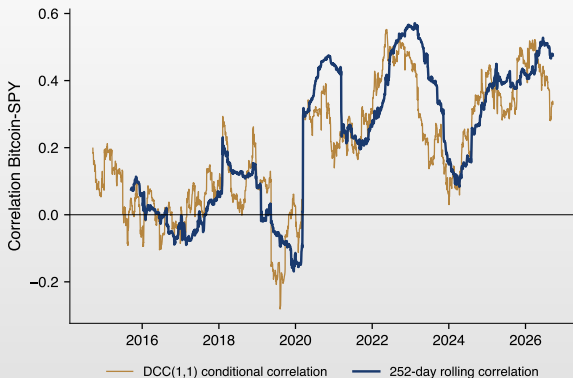
- Weight in SPY of the daily minimum-variance SPY-TLT portfolio, $w_t = (\sigma_{2t}^2 - \sigma_{12,t}) / (\sigma_{1t}^2 + \sigma_{2t}^2 - 2\sigma_{12,t})$, truncated to $[0, 1]$; same GARCH volatilities, correlation from DCC or constant (-0.30)
- Annualised volatility: 60/40 fixed 10.99%, GARCH with constant correlation 9.10%, GARCH with DCC 9.06%
- Most of the gain comes from GARCH volatilities; DCC adds little. Mean absolute weight gap DCC minus constant 0.019, largest 0.22 (8 June 2026)
- In-sample illustration: a fair test re-estimates on expanding windows and adds transaction costs (Chapter 8)

Dynamic Hedging Out of Sample: BET with Euro Stoxx 50



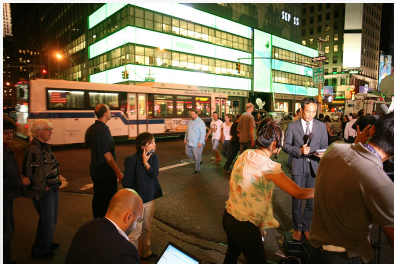
- Daily BET and Euro Stoxx 50 (almost synchronous closes) from 5 January 2010; estimation to end-2019 ($n = 2491$), evaluation 2020–2026 ($n = 1663$) with frozen parameters
- Static hedge: slope in the estimation window, $h = 0.337$; DCC hedge $h_t = \rho_t \sigma_{S,t} / \sigma_{F,t}$ ($\hat{a} = 0.009$, $\hat{b} = 0.964$): mean 0.29, range 0.13 to 0.84
- Variance removed in 2020–2026: static 24.7%, DCC 26.2%; the best constant hedge chosen ex post ($h = 0.41$): 25.6%
- Diebold & Mariano (1995) test on squared hedged returns (static minus DCC), HAC: $t = 1.29$, one-sided $p = 0.098$: a small gain, not significant at 5%

Bitcoin and US Equities: DCC Correlation



- Bitcoin and SPY on common trading days since 18 September 2014 ($T = 3\,018$); weekend Bitcoin moves enter the Monday return
- Average DCC correlation 0.04 before 2020, 0.31 since January 2020; peak 0.55 on 19 May 2022
- Very persistent: $a = 0.015$, $b = 0.982$, $a + b = 0.9968$; LR 105.6, bootstrap p under CCC = 0.005: CCC rejected
- From diversifier to risk asset: the full-sample correlation (0.24) hides two regimes; [Pele et al. \(2023\)](#) ask whether cryptos have become an asset class of their own

Contagion or Interdependence?



Lehman Brothers headquarters, New York, 15 September 2008, the day of the bankruptcy filing

Photo: Robert Scoble (2008); CC BY 2.0; Wikimedia

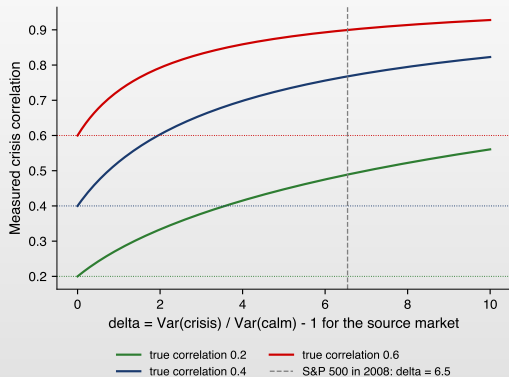
[Commons](#)

- Common finding: correlations measured during crises are higher than in calm periods; a high crisis correlation is not yet contagion, it must exceed the calm one
- Two very different explanations
 - ▶ **Contagion**: the transmission mechanism itself changes
 - ▶ **Volatility bias**: the mechanism is unchanged, but a more volatile source market mechanically raises the correlation
- **Forbes & Rigobon (2002)**: after correcting the bias, most of the 1987, 1994 and 1997 “contagion” became interdependence
- **Longin & Solnik (2001), Ang & Chen (2002)**: correlations rise in bear markets, not in bull markets, an asymmetry the Normal distribution cannot produce

The Heteroskedasticity Bias: Derivation

- Model: $y_t = \alpha + \beta x_t + \varepsilon_t$, $\text{Cov}(x_t, \varepsilon_t) = 0$, β and σ_ε^2 constant
- Correlation: $\rho^2 = \frac{\beta^2 \sigma_x^2}{\beta^2 \sigma_x^2 + \sigma_\varepsilon^2} = \left(1 + \frac{\sigma_\varepsilon^2}{\beta^2 \sigma_x^2}\right)^{-1}$
- Crisis: $\sigma_{x,c}^2 = (1 + \delta) \sigma_x^2$ with $\delta > 0$; nothing else changes
- Then $\rho_c = \rho \sqrt{\frac{1 + \delta}{1 + \delta \rho^2}}$, so $|\rho_c| > |\rho|$ for $0 < |\rho| < 1$: the measured correlation rises in absolute value although β is the same
- Inverting: $\rho^* = \frac{\rho_c}{\sqrt{1 + \delta(1 - \rho_c^2)}}$ is the crisis correlation adjusted for the volatility increase

How Much Can Volatility Alone Raise a Correlation?



- Measured crisis correlation ρ_c as a function of δ , for three true correlations that do not change
- Vertical line: the variance of 2-day S&P 500 returns in the 2008 crisis window was 7.5 times its calm-year level ($\delta = 6.54$)
- At this δ an unchanged true correlation of 0.4 is measured as 0.77, and 0.2 as 0.49

The Forbes–Rigobon Test and Its Inference

□ Procedure

- ▶ Choose the source market x , a calm window and a crisis window; use 2-day returns against time zones
- ▶ Compute $\delta = \hat{\sigma}_{x,c}^2 / \hat{\sigma}_{x,calm}^2 - 1$ and ρ^* ; test $H_0 : \rho^* \leq \rho_{calm}$ with Fisher z statistics, the convention of Forbes & Rigobon (2002)

□ Inference

- ▶ With δ fixed, $\text{Var}(\text{artanh } \rho^*) \approx 1/[n_c(1 + \delta(1 - \rho_c^2))] < 1/n_c$: the Fisher test is conservative
- ▶ δ is estimated and 2-day returns overlap: resample the calm and crisis windows jointly (stationary bootstrap, mean block 10 days, $B = 1999$) and test $D = \rho^* - \rho_{calm}$

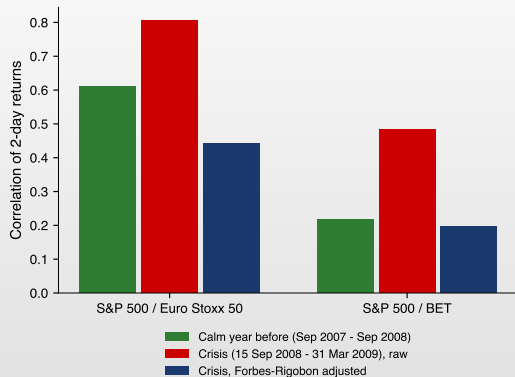
Assumptions to remember

- No omitted common shock: a global factor hitting both markets also raises σ_x and the adjustment over-corrects (Corsetti, Pericoli & Sbracia (2005); Appendix)
- No feedback from y to x (no endogeneity)
- The choice of windows matters: results can flip with a few weeks
- The correction goes too far if the mechanism really did change

Worked Example: The 2008 Crisis Step by Step

- ▣ Source market: S&P 500; calm year: 14 September 2007 – 12 September 2008 ($n = 242$ 2-day returns); crisis: 15 September 2008 – 31 March 2009 ($n = 130$)
- ▣ Variance ratio: $1 + \delta = 7.54$, so $\delta = 6.54$
- ▣ Euro Stoxx 50: $\rho_{calm} = 0.610$, $\rho_c = 0.804$
 - ▶ $\rho^* = 0.804 / \sqrt{1 + 6.54(1 - 0.804^2)} = 0.442$
 - ▶ One-sided p -value: 0.005 with ρ_c , 0.932 with ρ^*
- ▣ BET: $\rho_{calm} = 0.216$, $\rho_c = 0.485$, $\rho^* = 0.198$
 - ▶ One-sided p -value: 0.024 with ρ_c , 0.548 with ρ^*

2008: Raw versus Adjusted Crisis Correlations

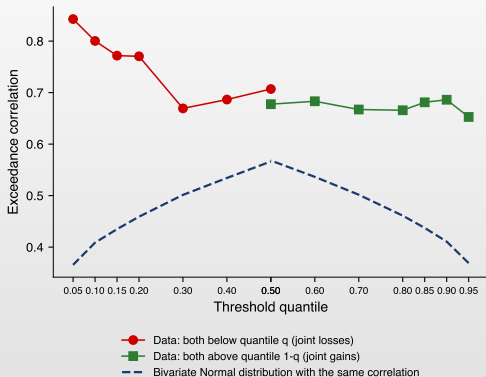


- Correlation of 2-day returns with the S&P 500: calm year, crisis (raw) and after the Forbes–Rigobon adjustment
- Raw correlations jump, Euro Stoxx 50 from 0.61 to 0.80, BET from 0.22 to 0.48; both rises significant at 5%, neither survives the adjustment
- With δ estimated: $\hat{\delta} = 6.54$, 90% interval [3.9, 9.6]; BET ρ^* in [0.16, 0.24], $p = 0.62$; Euro Stoxx 50 in [0.38, 0.52], $p = 0.98$
- Verdict: no evidence of an increase once volatility is adjusted for, under the maintained assumptions; 2008 was a global common shock, so the assumption is doubtful and the test has low power (Seminar B4 repeats it for 2020)

Exceedance Correlations

- ▣ Longin & Solnik (2001): correlation conditional on both returns being beyond a threshold
- ▣ Lower tail at quantile q : $\rho^-(q) = \text{Corr}(x, y \mid x \leq F_x^{-1}(q), y \leq F_y^{-1}(q))$
- ▣ Upper tail: the same with $x \geq F_x^{-1}(1 - q), y \geq F_y^{-1}(1 - q)$
- ▣ Benchmark: for a bivariate Normal distribution, $\rho^\pm(q) \rightarrow 0$ as $q \rightarrow 0$
 - ▶ Conditioning on extremes lowers the correlation even when nothing changes: compare with the Normal curve, not with ρ

Exceedance Correlations: S&P 500 and Euro Stoxx 50



- Weekly returns 2000–2026 ($n = 1\,393$, overall $\hat{\rho} = 0.78$); dashed: the bivariate Normal distribution with the same correlation (2 million simulated pairs)
- At $q = 0.10$: joint losses $\rho^- = 0.80$ (86 weeks), joint gains $\rho^+ = 0.69$ (70 weeks); Normal benchmark 0.41
- Both tails lie far above the Normal benchmark, and crashes are shared more than rallies
- A single coefficient cannot describe this: we need a model of the whole dependence structure, a copula

Why Correlation Is Not Enough

- ▣ Embrechts, McNeil & Straumann (2002): three fallacies about linear correlation
- ▣ Correlation measures only *linear* dependence and is not invariant to monotone transformations:
 $\text{Corr}(x, y) \neq \text{Corr}(e^x, e^y)$
- ▣ Margins and correlation do not determine the joint distribution
 - ▶ Two portfolios with identical margins and ρ can have very different Value-at-Risk (Chapter 7)
- ▣ Attainable correlations: for given margins, $\rho \in [\rho_{\min}, \rho_{\max}]$, reached by countermonotone and comonotone pairs
 - ▶ Lognormal $X = e^{\sigma_1 Z}$, $Y = e^{\pm \sigma_2 Z}$: $\rho_{\max} = \frac{e^{\sigma_1 \sigma_2} - 1}{\sqrt{(e^{\sigma_1^2} - 1)(e^{\sigma_2^2} - 1)}}$, $\rho_{\min}$ with $e^{-\sigma_1 \sigma_2}$ in the numerator
 - ▶ $\sigma_1 = 1$, $\sigma_2 = 3$: $\rho_{\max} = 0.162$ and $\rho_{\min} = -0.008$: perfectly dependent, yet almost uncorrelated

Sklar's Theorem: Margins plus Copula

- ▣ A **copula** is a joint distribution function on $[0, 1]^d$ with uniform margins
- ▣ Sklar (1959): for any joint distribution F with margins $F_1, \dots, F_d$ there is a copula C with $F(x_1, \dots, x_d) = C(F_1(x_1), \dots, F_d(x_d))$
 - ▶ C is unique if the margins are continuous; conversely, any copula and any margins give a valid joint distribution
 - ▶ With densities: $f(x) = c(F_1(x_1), \dots, F_d(x_d)) \prod_i f_i(x_i)$ (the comonotone copula has none)
- ▣ Margins carry volatility, skewness and fat tails of each asset (Chapters 1 and 5); the copula carries all the dependence and is invariant to increasing transformations of each variable
- ▣ Three reference copulas and the Fréchet–Hoeffding bounds $W \leq C \leq M$ (Nelsen (2006))
 - ▶ Independence $\Pi(u, v) = uv$; comonotonicity $M(u, v) = \min(u, v)$; countermonotonicity $W(u, v) = \max(u + v - 1, 0)$

Rank Correlations: Kendall and Spearman

- **Kendall's tau** (Kendall (1938)): concordant minus discordant pairs
 - ▶ $\tau = P[(X_1 - X_2)(Y_1 - Y_2) > 0] - P[(X_1 - X_2)(Y_1 - Y_2) < 0]$
 - ▶ Copula form: $\tau = 4 \iint C \, dC - 1$
- **Spearman's rho** (Spearman (1904)): Pearson correlation of the ranks
 - ▶ Copula form: $\rho_S = 12 \iint (C(u, v) - uv) \, du \, dv$
- Both depend only on the copula; both equal 1 for M , 0 for Π , -1 for W

Links to copula parameters

Copula	τ
Gaussian, t	$\frac{2}{\pi} \arcsin \rho$
Clayton	$\theta / (\theta + 2)$
Gumbel	$1 - 1/\theta$
Frank	$1 - \frac{4}{\theta} [1 - D_1(\theta)]$

- $D_1(\theta) = \frac{1}{\theta} \int_0^\theta \frac{t}{e^t - 1} dt$ (Debye function)
- Gaussian: $\rho_S = \frac{6}{\pi} \arcsin(\rho/2)$

Five Copula Families

Family	$C(u, v)$	Parameter
Gaussian	$\Phi_{2,\rho}(\Phi^{-1}(u), \Phi^{-1}(v))$	$-1 < \rho < 1$
Student t (Demarta & McNeil (2005))	$t_{2,\rho,\nu}(t_\nu^{-1}(u), t_\nu^{-1}(v))$	$-1 < \rho < 1, \nu > 0$
Clayton (Clayton (1978))	$(u^{-\theta} + v^{-\theta} - 1)^{-1/\theta}$	$\theta > 0$
Gumbel (Gumbel (1960))	$\exp(-[(-\ln u)^\theta + (-\ln v)^\theta]^{1/\theta})$	$\theta \geq 1$
Frank (Frank (1979))	$-\frac{1}{\theta} \ln\left(1 + \frac{(e^{-\theta u} - 1)(e^{-\theta v} - 1)}{e^{-\theta} - 1}\right)$	$\theta \neq 0$

- Elliptical (Gaussian, t): extend to d dimensions with a correlation matrix, the standard choice in risk systems; $\nu \rightarrow \infty$ turns the t copula into the Gaussian one
- Archimedean (Clayton, Gumbel, Frank): $C(u, v) = \varphi^{-1}(\varphi(u) + \varphi(v))$ from a generator φ ; one parameter, closed form, easy simulation, but exchangeable and poor in high dimension
- Rotation by 180° (the survival copula of $(1 - U, 1 - V)$) moves the tail dependence to the opposite corner (densities and generators in the Appendix)

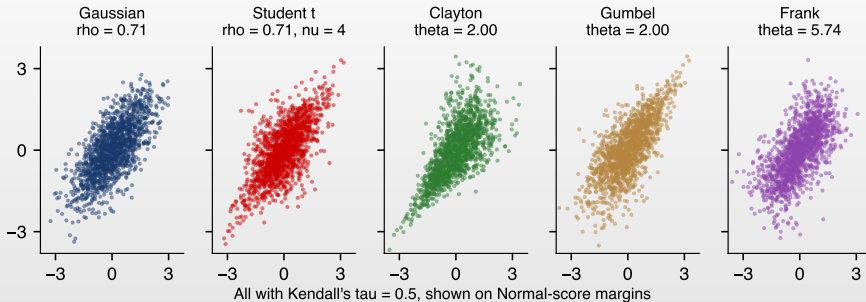
Tail Dependence

- Lower tail:
 $\lambda_L = \lim_{q \downarrow 0} P(V \leq q \mid U \leq q) = \lim_{q \downarrow 0} C(q, q)/q$, with
 $U = F_1(X_1)$, $V = F_2(X_2)$
- Upper tail: $\lambda_U = \lim_{q \downarrow 0} P(V > 1 - q \mid U > 1 - q)$
- $\lambda > 0$: extremes arrive together no matter how extreme;
 $\lambda = 0$: asymptotic independence
- Tail dependence is a copula property, invisible in ρ ; at a finite q , $C(q, q)/q$ is a probability, not yet the limit

Closed forms

	λ_L	λ_U
Gaussian	0	0
Student t	$2 t_{\nu+1} \left(-\sqrt{\frac{(\nu+1)(1-\rho)}{1+\rho}} \right)$	same
Clayton	$2^{-1/\theta}$	0
Gumbel	0	$2 - 2^{1/\theta}$
Frank	0	0

Five Copulas with the Same Kendall's Tau

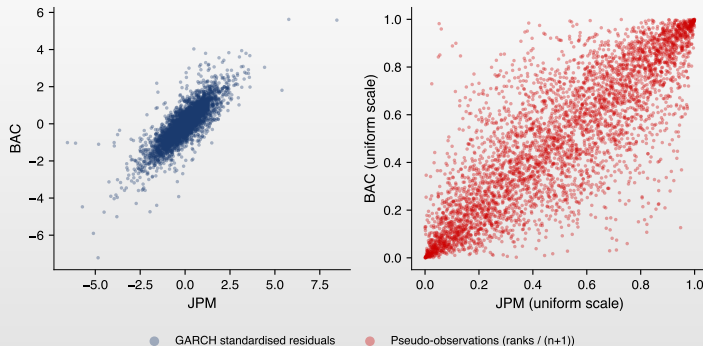


- 1 500 simulated pairs from each copula with $\tau = 0.5$ on Normal-score margins $(\Phi^{-1}(u), \Phi^{-1}(v))$; Gaussian and t $\rho = 0.707$, Clayton and Gumbel $\theta = 2.0$, Frank $\theta = 5.74$
- Corners: Gaussian and Frank thin out, t stays dense in both, Clayton is sharp lower-left (joint crashes), Gumbel sharp upper-right
- Equal Kendall dependence does not determine joint-tail risk

Estimating Copulas

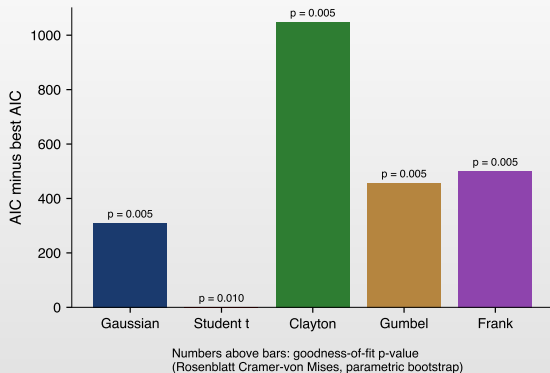
- ▣ **Full maximum likelihood**: margins and copula estimated jointly; efficient but computationally demanding and sensitive to misspecification
- ▣ **IFM** (inference functions for margins, [Joe \(2005\)](#)): two steps
 - ▶ Fit parametric margins (e.g. GARCH with Student t); set $\hat{u}_{it} = \hat{F}_i(x_{it})$; maximise $\sum_t \ln c(\hat{u}_{1t}, \hat{u}_{2t}; \theta)$
- ▣ **CML** (canonical maximum likelihood, [Genest, Ghoudi & Rivest \(1995\)](#)): no parametric margins
 - ▶ Pseudo-observations $\hat{u}_{it} = \text{rank}(x_{it}) / (T + 1)$; then maximise the copula likelihood
 - ▶ Robust to misspecified margins; the $T + 1$ keeps $\hat{u}$ away from 0 and 1
- ▣ For returns: filter each series with GARCH first, then use the ranks of the standardised residuals
- ▣ Model choice: $\text{AIC} = 2k - 2\ell$, $\text{BIC} = k \ln T - 2\ell$

From Residuals to Pseudo-Observations: JPM and BAC



- JPMorgan Chase (JPM) and Bank of America (BAC), daily 2010–2026 ($T = 4\,202$), GARCH(1,1)-filtered
- Kendall's $\hat{\tau} = 0.641$, Spearman's $\hat{\rho}_S = 0.822$
- Both corners of the uniform scatter are dense: joint crashes and joint rallies

Copula Estimates and Goodness of Fit: JPM and BAC



	$\hat{\theta}$ or $\hat{\rho}$	$\hat{\nu}$	λ_L	λ_U
Gaussian	0.833	–	0.000	0.000
Student t	0.842	4.24	0.531	0.531
Clayton	2.384	–	0.748	0.000
Gumbel	2.587	–	0.000	0.693
Frank	9.152	–	0.000	0.000

- Bars: AIC distance to the best copula; numbers: bootstrap goodness-of-fit p -values (full table in the Appendix)
- Student t wins by far: $\hat{\nu} = 4.2$, $\hat{\lambda} = 0.53$ in both tails; Gaussian beats the Archimedean ones: strong, roughly symmetric dependence
- Every family is rejected at 5%; the t copula has by far the smallest statistic (0.082 against 0.968 for the Gaussian)

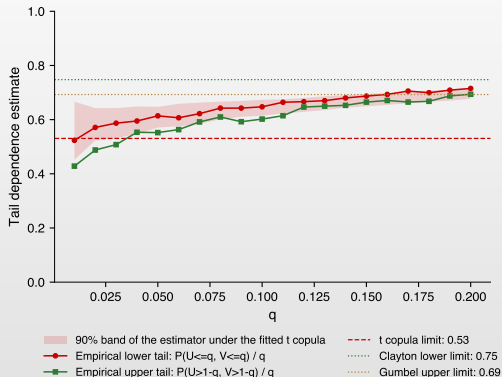
Inference for Copula Parameters

- **Inversion of $\hat{\tau}$:** a U-statistic with a Hoeffding-projection standard error; $\hat{\theta} = g(\hat{\tau})$ by the delta method (Genest & Rivest (1993))
 - ▶ Elliptical copulas: $\tau = \frac{2}{\pi} \arcsin \rho$ for every ν , so τ identifies ρ , never ν
 - ▶ JPM-BAC: $\hat{\tau} = 0.641$ (se 0.0065), $\hat{\rho}_{\tau} = \sin(\pi \hat{\tau}/2) = 0.845$ (se 0.0055); CML $\hat{\rho} = 0.842$ (se 0.0053)
- **CML: the ranks are estimated**, so the inverse Hessian is too small; the Genest, Ghoudi & Rivest (1995) sandwich adds the rank terms
 - ▶ Increase of the se: +60% for the Gaussian ρ (S&P 500–Euro Stoxx 50 weekly), +17% for the t-copula ρ and +1% for ν (JPM-BAC)
- **After GARCH filtering** (Chen & Fan (2006)): CML on the ranks of estimated residuals has the same limit as on the true innovations
 - ▶ Conditions: i.i.d. innovations with copula C_{θ} and a correctly specified GARCH; then the sandwich stays valid and, unlike DCC, no step-1 correction is needed (Monte Carlo in the Appendix)

Goodness of Fit: The Rosenblatt Test

- If $(U, V) \sim C_\theta$, then $E_1 = U$ and $E_2 = h(V | U) = \partial C_\theta(U, V) / \partial u$ are independent uniforms
- Statistic (Genest, Rémillard & Beaudoin (2009)): $S_n = n \int_{[0,1]^2} (D_n(e) - e_1 e_2)^2 de$, D_n the empirical distribution of (E_1, E_2)
- p -value by parametric bootstrap, because θ is estimated
 - ▶ Simulate n pairs from $C_{\hat{\theta}}$, re-rank them into pseudo-observations, re-estimate, recompute $S_n^{(b)}$
 - ▶ $p = (1 + \#\{S_n^{(b)} \geq S_n\}) / (B + 1)$; for JPM-BAC $B = 200$ (100 for Student t)
- On GARCH residuals, Rémillard (2017) shows that rank-based tests keep their limit and the parametric bootstrap applies
- With $T = 4202$ the test has high power against small departures: a rejection does not say which feature is missing

Empirical Tail Dependence: JPM and BAC



- $\hat{\lambda}_L(q) = \hat{C}(q, q)/q$ and $\hat{\lambda}_U(q)$ for q from 1% to 20%; band: 90% range of the same estimator on data simulated from the fitted t copula
- $q = 5\%$: $\hat{\lambda}_L = 0.61$, $\hat{\lambda}_U = 0.55$; $q = 1\%$: 0.52 and 0.43
- When JPM has one of its 5% worst days, BAC is in its own worst 5% about 61% of the time
- Lower tail inside the t-copula band (0.57–0.65 at 5%): the t copula describes the crashes well
- Finite- q values are threshold probabilities, not the limit λ ; few points lie in the far corner

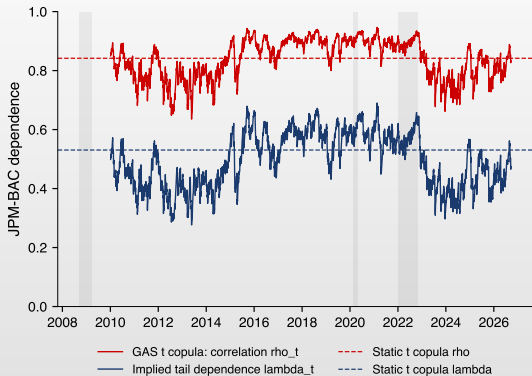
Is the Tail Asymmetry Real?

- Schmidt & Stadtmüller (2006): $\hat{\lambda}_L = \frac{n}{k} \hat{C}_n(\frac{k}{n}, \frac{k}{n})$, the chart estimator with $q = k/n$; a larger k lowers the variance but raises the bias (conditions in the Appendix)
- $H_0 : \lambda_L(q) = \lambda_U(q)$ at a fixed q (threshold probabilities, not the limits); bootstrap of residual pairs, re-ranked, $B = 999$

q	k	$\hat{\lambda}_L - \hat{\lambda}_U$	95% interval, i.i.d.	95% interval, blocks
5%	210	0.062	$[-0.014, 0.148]$	$[-0.014, 0.138]$
1%	42	0.095	$[-0.095, 0.286]$	$[-0.119, 0.286]$

- The block version (stationary bootstrap, mean block 1.6 and 1.0 days, Politis & White (2004) rule) allows serial dependence in the tail indicators; the interval at 1% is wider, as the bias–variance trade-off predicts
- All intervals contain zero: the asymmetry is not significant, so the symmetric t copula is not contradicted in the tails

A Dynamic t Copula for JPM and BAC

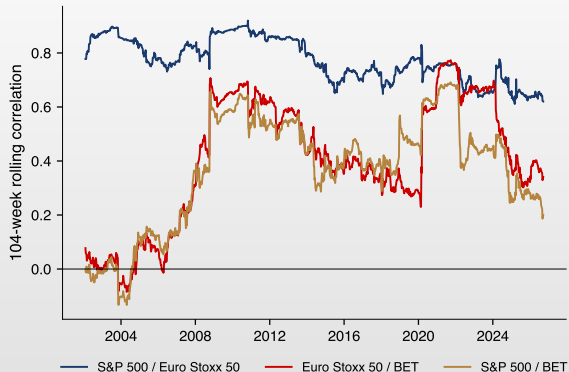


- GAS (generalised autoregressive score) t copula, [Creal, Koopman & Lucas \(2013\)](#): $\rho_t = \tanh f_t$, $f_{t+1} = \omega + A s_t + B f_t$, $s_t = \partial \ln c(\hat{u}_t, \hat{v}_t; \rho_t, \nu) / \partial f_t$ (unit scaling), same pseudo-observations
- $A = 0.025$, $B = 0.9926$, $\nu = 5.35$ (static 4.24); likelihood ratio against the static t copula 245.5
- ρ_t from 0.64 (23 May 2013) to 0.95 (18 February 2021); λ_t from 0.28 to 0.69 around a static 0.53
- Once ρ moves, a larger ν suffices: part of the static fat tails was time variation in dependence
- Under $A = 0$, B is not identified: judge the LR by bootstrap, not by χ^2 ; time variation is one explanation consistent with the rejection of every static family (Seminar B6)

A Cautionary Tale: The Gaussian Copula and 2008

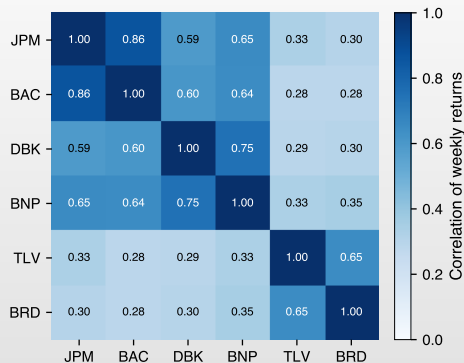
- ▣ Li (2000) modelled joint default times with a Gaussian copula: one correlation parameter, calibrated to market spreads, priced whole credit portfolios and became the standard for tranches of collateralised debt obligations (CDO)
- ▣ Zero asymptotic tail dependence: for the most extreme joint defaults it gives lower probabilities than tail-dependent models calibrated to the same data
- ▣ A static correlation calibrated in calm years: exactly the Forbes–Rigobon and DCC lessons of this chapter

US, Euro-Area and Romanian Equities, 2000–2026



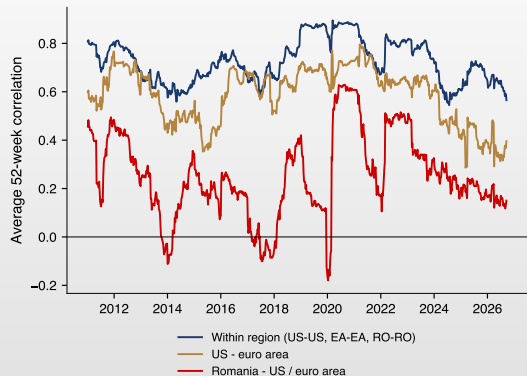
- Weekly returns, 104-week window, common trading days, Friday closes
- Euro Stoxx 50–BET: 0.33 in 2000–2009, 0.45 in 2010–2019, 0.61 since 2020; S&P 500–Euro Stoxx 50: 0.84, 0.78, 0.70
- Rolling Euro Stoxx 50–BET: 0.31 at end-2007, 0.65 at end-2009, 0.29 at end-2019, 0.76 at end-2021, 0.35 now
- Jumps in crises and partial reversal afterwards: integration or the Forbes–Rigobon volatility bias? (Seminar C1)

Banks: Correlations of Weekly Returns, 2010–2026



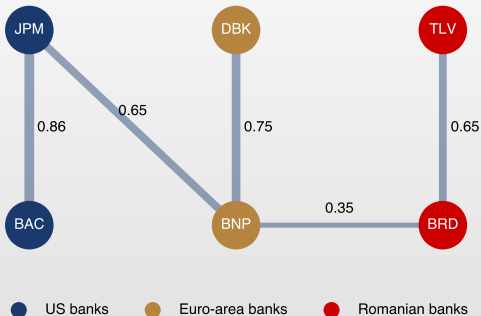
- US: JPM and BAC; euro area: DBK (Deutsche Bank) and BNP (BNP Paribas); Romania: TLV (Banca Transilvania) and BRD (BRD-Groupe Société Générale); 871 weeks from 15 January 2010
- Three regional blocks: US 0.86, euro area 0.75, Romania 0.65
- Cross-region links are weakest for Romania: BNP–TLV 0.33, JPM–BRD 0.30
- Bank returns share country-level factors: sovereign risk, domestic rates, regulation

Bank Correlations over Time



- Average pairwise 52-week correlation of weekly returns: within regions, between the US and the euro area, between Romania and the others
- Full period: within regions 0.73, US–euro area 0.59, Romania–others 0.25
- March–June 2020: 0.87, 0.69 and 0.56: the Romanian link more than doubled in a global shock; latest 0.57, 0.40, 0.15
- Diversification across banking systems works in normal times and weakens exactly in systemic events

A Correlation Network of Banks

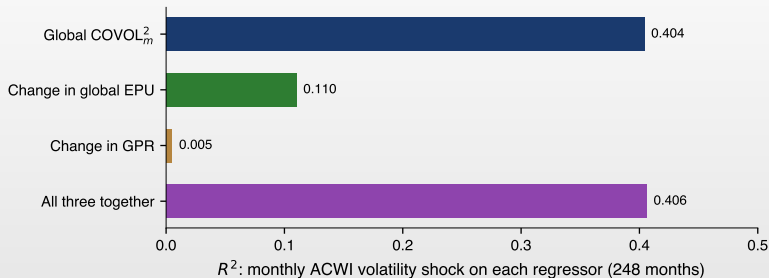


- Minimum spanning tree (Mantegna (1999)) on $d_{ij} = \sqrt{2(1 - \rho_{ij})}$: the 5 strongest links that connect all 6 banks without cycles
- Edges: JPM-BAC (0.86); JPM-BNP (0.65); DBK-BNP (0.75); BNP-BRD (0.35); TLV-BRD (0.65); Romania is attached through BNP-BRD
- Chapter 4 uses $\sqrt{(1 - \rho)/2}$: a factor 2 apart, both increase with $1 - \rho$, so the tree is the same
- The tree is undirected and contemporaneous: it does not identify transmission paths, only hypotheses for the network measures of Chapter 18 (literature in the Appendix)

Case Study: Question and Model

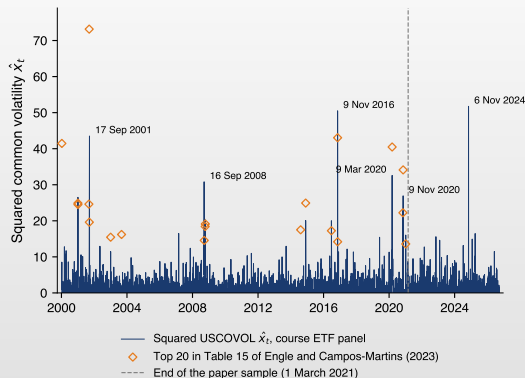
- Engle & Campos-Martins (2023): which events move the volatilities of almost all countries and asset classes on the same day, and how large is this common shock? Its daily size is global COVOL (common volatility)
- The distinct idea: a correct DCC gives residuals e_t with $E_{t-1}(e_t e_t^\top) = I$, but uncorrelated is not independent: the squares can comove, $\Psi_t = E_{t-1}[(e_t^2 - \iota)(e_t^2 - \iota)^\top]$ has non-zero off-diagonal elements
 - ▶ The cross-sectional analogue of ARCH: correlated squares across assets on one day, not over time; [Herskovic, Kelly, Lustig & Van Nieuwerburgh \(2016\)](#) document such a factor in the idiosyncratic volatility of US stocks
- Model: $e_{it} = \sqrt{s_i x_t + 1 - s_i} \varepsilon_{it}$, ε_{it} i.i.d. $\mathcal{N}(0, 1)$, $x_t > 0$, $E_{t-1}(x_t) = 1$, loading $s_i \in [0, 1]$; x_t is the squared global COVOL
 - ▶ Equal loadings $s_i = 1$: $\hat{x}_t = \frac{1}{N} \sum_i e_{it}^2$; unequal loadings: alternating maximum likelihood over x_t (one cross-section per day) and s_i (one time series per asset), with $s^\top s = 1$ and mean $x_t = 1$
 - ▶ Test ξ of $H_0: \Psi_{ij} = 0, i \neq j$; data: 47 country ETFs listed on the NYSE, 2000–2021, and 23 US asset-class series (details in the Appendix)

Case Study: Realised Shocks against News Indices



- Monthly, 248 months (Table 8 of [Engle & Campos-Martins \(2023\)](#)): R^2 of the ACWI (MSCI All Country World Index) volatility shock on COVOL $_m^2$ is 0.404, against 0.110 for the change in EPU ([Baker, Bloom & Davis \(2016\)](#)) and 0.005 for GPR ([Caldara & Iacoviello \(2022\)](#)); adding both news indices raises R^2 only to 0.406
- Top days include 9 March 2020 ($\hat{x}_t = 51.4$), Brexit and a day with a positive mean return: COVOL measures the size of the common shock, not its sign; news indices measure what people fear, COVOL the shocks that hit prices (event table in the Appendix)

Case Study: Asset-Class COVOL on the Course ETFs, 2000–2026



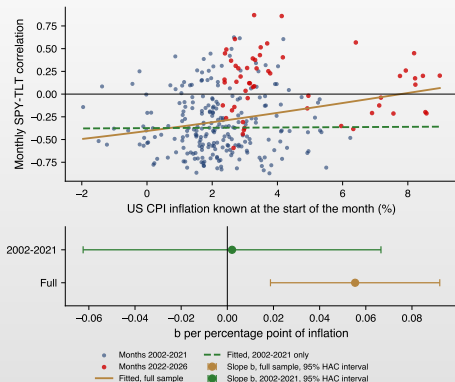
- 20 US-listed ETFs and their PC1 (first principal component), 4 January 2000 – 18 September 2026, 6 717 days; AR(1) mean with PC1, GARCH(1,1), then the alternating estimation (13 iterations); diamonds: the 20 largest values of Table 15 of [Engle & Campos-Martins \(2023\)](#)
- Top days up to 1 March 2021: 9 November 2016 ($\hat{x}_t = 50.5$), 17 September 2001 (43.5), 9 March 2020 (32.6): the 2016 US election, the reopening after 9/11 and the COVID-19 crash
- 12 of the 20 largest days of the application before March 2021 are among the 20 dates of Table 15; with $s^\top s = 1$ and mean $\hat{x}_t = 1$ the scale depends on the panel, so compare rankings, not levels
- Largest in the full sample: 6 November 2024 (51.8, mean ETF return 0.95%), the day after the 2024 US election: a large common shock on an up day

 MFM_ch6_covol

An Open Question and the Discovery Loop

- Is the positive stock–bond correlation since 2022 a durable regime governed by inflation, and can it be forecast?
 - ▶ Campbell, Pflueger & Viceira (2020): the sign of the bond–stock comovement follows the inflation–output relation and the monetary-policy regime; Molenaar, Sénéchal, Swinkels & Wang (2024): evidence across countries and a long history
 - ▶ This chapter: daily SPY–TLT correlation -0.40 in 2002–2021 and 0.11 in 2022–2026; DCC above its target on 92% of days since 2022
- Hypothesis with monthly realised correlations ρ_m : $z_m = \text{artanh } \rho_m = a + b \pi_{m-2} + u_m$, π the annual US CPI (consumer price index) inflation known when month m starts
 - ▶ H_0 : $b = 0$; H_1 : $b > 0$; HAC t-statistic; falsified if $b \approx 0$ in periods where inflation varied but the regime did not
- The loop: an AI assistant drafts the literature map, competing hypotheses, the code, a robustness plan and a referee critique (prompts in the Appendix)
 - ▶ The researcher resolves every DOI, replicates the chapter's numbers first and fixes the plan before the first run

Mini-Case: Does Inflation Drive the Stock–Bond Correlation?



- Monthly SPY-TLT correlation from daily returns against the inflation known at the start of the month (π_{m-2} : CPI for month m is published in month $m+1$); current CPI vintage, so a retrospective test, not a real-time one
- Full sample, 287 months: $\hat{b} = 0.055$ per percentage point, HAC se 0.019, $t = 2.96$, $R^2 = 0.047$; mean $\rho = -0.12$ when inflation exceeds 3% (89 months), -0.30 otherwise
- 2002–2021 only (232 months, 53 with inflation above 3%): $\hat{b} = 0.002$, $t = 0.06$: the full-sample slope rests on the post-2022 episode
- Associational, and one regime change cannot separate the inflation level from the policy regime (Campbell, Pflueger & Viceira (2020)): the claim is not yet established

What the Researcher Checks, and a Project Seed

Checks before any claim

- Look-ahead: release calendar and data vintages (the lag handles publication delay, not revisions)
- Calendars: join prices on common days before computing returns
- Persistent regressor, few regimes: subsamples, long HAC lags, a placebo with simulated persistent series
- Volatility bias: control for realised volatility; multiple testing: fix measures and windows in advance
- Invented references: an unresolvable DOI is dropped and logged in `AI_ERRORS.md`

Your turn: a team-project seed

- Does a DCC whose target depends on inflation and the policy rate forecast the SPY-TLT correlation better out of sample than the plain DCC?
- Replicate first: $a + b = 0.986$, $\bar{Q}_{12} = -0.25$ and the slope $\hat{b} = 0.055$
- Deliver a one-page pre-registration, code on data/market plus CPI, a Diebold-Mariano comparison and a referee-style self-critique; log AI use in `AI_USE.md`

Decision Checklist: Question, Estimator, Uncertainty, Failure

Question	Estimator	Uncertainty	Fails when
Did a correlation change?	Pearson on joined prices	HAC delta method, block bootstrap	Fisher under clustering; daily data across time zones
Covariance for risk today?	EWMA, DCC	bootstrap of both steps	the target $\bar{Q}$ moves with the regime
When did it break?	Wied–Krämer–Dehling on GARCH residuals	Brownian-bridge limit	the date is read off a chart
Contagion?	Forbes–Rigobon ρ^*	stationary bootstrap, δ estimated	a common shock or feedback
Joint extremes?	copula on GARCH pseudo-observations	rank sandwich, parametric bootstrap	Gaussian copula ($\lambda = 0$), static copula when dependence moves

□ Next: Chapter 7 turns DCC covariances and simulated copulas into portfolio VaR and ES

Appendix: Further Study, Code and Data

Advanced courses on quantinar.com

- ☐  Measuring Statistical Risk
- ☐  Multivariate Statistical Analysis
- ☐  Minimum Spanning Tree
- ☐  FRM: Financial Risk Meter

Code, data and textbooks

- ☐ All charts and tables: github.com/danpele/MFM/tree/main/Quantlets/Ch_06
- ☐ Textbooks: Franke, Härdle & Hafner (2019); Härdle & Simar (2019); Nelsen (2006)

Appendix: Inference on a Correlation, the Delta Method

- ▣ $\hat{\rho} = g(\hat{m})$, with $\hat{m}$ the sample mean of $m_t = (x_t, y_t, x_t^2, y_t^2, x_t y_t)$
- ▣ Delta method: $\sqrt{T}(\hat{\rho} - \rho) \rightarrow N(0, \nabla g^\top \Omega \nabla g)$, Ω the long-run covariance of m_t ; requires stationarity, finite fourth moments of (x_t, y_t) and weak dependence (mixing)
 - ▶ $\hat{\Omega}$: HAC estimator with Bartlett kernel and L lags (Newey & West (1987))
 - ▶ Equivalent influence-function form: $\hat{\rho} - \rho \approx \frac{1}{T} \sum_t \psi_t$, $\psi_t = \tilde{x}_t \tilde{y}_t - \frac{\rho}{2}(\tilde{x}_t^2 + \tilde{y}_t^2)$
- ▣ i.i.d. Normal pairs: $\text{Var}(\psi_t) = (1 - \rho^2)^2$, the Fisher variance; fat tails typically raise $\text{Var}(\psi_t)$ and GARCH can make ψ_t autocorrelated; with infinite fourth moments the Normal limit fails
- ▣ Interval on the Fisher scale: $\tanh(\text{artanh } \hat{\rho} \pm 1.96 \hat{\text{se}} / (1 - \hat{\rho}^2))$
- ▣ Last SPY–TLT window (17 September 2026): $\hat{\rho} = 0.25$, Fisher $[0.13, 0.36]$, HAC $[0.07, 0.41]$; largest width ratio 2.02

Appendix: The VEC and BEKK Models

- **VEC** (Bollerslev, Engle & Wooldridge (1988)): every element of H_t depends on all squares and cross-products
 - ▶ $\text{vech}(H_t) = c + A \text{vech}(r_{t-1} r_{t-1}^\top) + B \text{vech}(H_{t-1})$
 - ▶ Positive definiteness is not guaranteed; parameters: $N(N+1)/2 \cdot [1 + N(N+1)]$
- **BEKK** (Engle & Kroner (1995)): positive definite by construction if $CC^\top$ is positive definite
 - ▶ $H_t = CC^\top + A^\top r_{t-1} r_{t-1}^\top A + B^\top H_{t-1} B$
 - ▶ Parameters: $N(N+1)/2 + 2N^2$; hard to interpret, and the likelihood is flat in high dimension
- DCC keeps univariate GARCH for each variance and adds two correlation parameters: $3N + 2$ in total

Appendix: Two-Step Estimators, the Correct Covariance

- Stack the estimating equations: $\sum_t s_{1t}(\theta_1) = 0$ (step 1, GARCH) and $\sum_t s_{2t}(\theta_1, \theta_2) = 0$ (step 2, correlation)
- $A_{jk} = \mathbb{E}[\partial s_{jt} / \partial \theta_k^\top]$: A_{21} measures how the correlation score reacts to the GARCH parameters
- Newey & McFadden (1994), Section 6: $\sqrt{T}(\hat{\theta}_2 - \theta_2) \rightarrow N(0, V_2)$ with
 - ▶ $V_2 = A_{22}^{-1} \mathbb{E}[(s_{2t} - A_{21}A_{11}^{-1}s_{1t})(s_{2t} - A_{21}A_{11}^{-1}s_{1t})^\top] A_{22}^{-1\top}$
- Step-2 standard errors keep only s_{2t} : they drop $A_{21}A_{11}^{-1}s_{1t}$, the GARCH estimation error carried into step 2
- With correct conditional moments the scores are martingale differences, so the middle term is their variance; otherwise use its long-run (HAC) version
 - ▶ With targeting, $\bar{Q}$ is estimated too: its moment equations belong to step 1 (θ_1 includes $\text{vech } \bar{Q}$)
- Engle & Sheppard (2001) derive V_2 for DCC; a simple alternative bootstraps the *whole* two-step procedure (next slide)

Appendix: Bootstrap of Both DCC Steps

- ▣ Parametric residual bootstrap, $B = 199$: decorrelate the residuals with the Cholesky factor of R_t , resample them and simulate DCC innovations with $(\hat{a}, \hat{b}, \tilde{Q})$
- ▣ Step 2 only: re-estimate (a, b) on the simulated innovations; both steps: rebuild returns with the fitted GARCH, re-estimate GARCH, $\tilde{Q}$ and (a, b) on the same innovations
- ▣ SPY-TLT: step 1 raises the se by 2% (a) and 1% (b); the step-2 Hessian $\text{se}(\hat{b})$ is 8% above the bootstrap se of both steps
- ▣ The CCC-null bootstrap of the LR test uses the same machinery with $a = b = 0$: constant-correlation innovations, simulated GARCH returns, both steps re-estimated

Appendix: Robert Engle and the DCC Model



Robert F. Engle at the Western Economic Association
International conference, Santiago, 2017

Photo: [Econterms \(2017\)](#); CC BY-SA 4.0; [Wikimedia](#)

[Commons](#)

- Nobel Memorial Prize in Economic Sciences 2003 for ARCH models (Chapter 5)
- [Engle \(2002\)](#): “Dynamic Conditional Correlation”
 - ▶ Keeps the flexibility of univariate GARCH for each variance
 - ▶ Adds a simple, estimable law of motion for correlations
- Refinement: the corrected DCC of [Aielli \(2013\)](#), which makes $\bar{Q}$ a consistent target

Appendix: Contagion Tests beyond Forbes–Rigobon

- ▣ Corsetti, Pericoli & Sbracia (2005): with a common factor the Forbes–Rigobon adjustment over-corrects, biasing the test towards “no contagion”; they give a factor-model version
- ▣ Rigobon (2003): identification through heteroskedasticity; the variance shift between regimes identifies the structural coefficient β , an instrumental-variables idea
- ▣ Bootstrap for an estimated δ : resample the calm and crisis windows with a stationary bootstrap, recompute $\hat{\delta}$, ρ^* and ρ_{calm} in each draw; p is the share of draws with $\rho^* - \rho_{calm} \leq 0$
- ▣ The 2008 verdict survives once the uncertainty of δ is included; the common-factor critique remains (Seminar A3, B4)

Appendix: Copula Densities and Constructions

- Gaussian density: $c = \frac{1}{\sqrt{1-\rho^2}} \exp\left(-\frac{\rho^2(x^2+y^2)-2\rho xy}{2(1-\rho^2)}\right)$, $x = \Phi^{-1}(u)$, $y = \Phi^{-1}(v)$; symmetric, no tail dependence for $|\rho| < 1$
- Student t: small ν means strong joint extremes, equal in both tails; the t copula is the building block of risk systems with a correlation matrix
- Archimedean generators φ in $C(u, v) = \varphi^{-1}(\varphi(u) + \varphi(v))$
 - Clayton $\varphi(t) = (t^{-\theta} - 1)/\theta$; Gumbel $\varphi(t) = (-\ln t)^\theta$; Frank $\varphi(t) = -\ln \frac{e^{-\theta t} - 1}{e^{-\theta} - 1}$
- Asymmetric tails: Clayton for joint crashes of returns, Gumbel for joint extremes of losses; the 180° rotation $\bar{C}(u, v) = u + v - 1 + C(1 - u, 1 - v)$ moves the tail to the opposite corner

Appendix: Copula Estimates for JPM and BAC, Full Table

Copula	$\hat{\theta}$ or $\hat{\rho}$	$\hat{\nu}$	ℓ	ΔAIC	λ_L	λ_U	fit p
Gaussian	0.833	—	2486.0	309.3	0.000	0.000	0.005
Student t	0.842	4.24	2641.7	0.0	0.531	0.531	0.010
Clayton	2.384	—	2116.7	1048.0	0.748	0.000	0.005
Gumbel	2.587	—	2412.2	456.9	0.000	0.693	0.005
Frank	9.152	—	2390.8	499.8	0.000	0.000	0.005

- CML on pseudo-observations of GARCH residuals; ΔAIC relative to the best model; goodness-of-fit p from the Rosenblatt test with parametric bootstrap

Appendix: Rank Estimators and Their Precision

- $\hat{\tau}$ is a U-statistic; its Hoeffding projection gives $\sqrt{n}(\hat{\tau} - \tau) \rightarrow N(0, 16\text{Var}[2C(U, V) - U - V])$
- Plug-in: $\widehat{\text{se}}(\hat{\tau}) = 4 \text{sd}(2\hat{C}_n(\hat{U}_i, \hat{V}_i) - \hat{U}_i - \hat{V}_i) / \sqrt{n}$; inversion estimators $\hat{\theta} = g(\hat{\tau})$ by the delta method (Genest & Rivest (1993))
- The moment estimator is almost as precise as CML for ρ and is a robust starting value for the likelihood
- CML sandwich (Genest, Ghoudi & Rivest (1995)): $\sqrt{n}(\hat{\theta} - \theta) \rightarrow N(0, A^{-1}\Sigma A^{-1})$, A the Fisher information of the copula, $\Sigma = \text{Var}[\dot{\ell}_{\theta}(U, V) + W_1(U) + W_2(V)]$
 - ▶ $W_1(u) = \iint 1\{u \leq u'\} \partial_u \dot{\ell}_{\theta}(u', v') dC(u', v')$: the price of ranking; W_2 likewise in v
 - ▶ Gaussian copula: variance $(1 - \rho^2)^2$ for CML, against $(1 - \rho^2)^2 / (1 + \rho^2)$ with known margins
- S&P 500–Euro Stoxx 50 weekly, Gaussian: se 0.0086 (inverse Hessian) versus 0.0138 (sandwich); JPM–BAC, t: ρ 0.0046 versus 0.0053, ν 0.359 versus 0.364

Appendix: Chen and Fan (2006), GARCH Filtering and the Limit

- Model: $r_{it} = \mu_i + \sigma_{it}\varepsilon_{it}$ with GARCH σ_{it} ; innovations $(\varepsilon_{1t}, \varepsilon_{2t})$ i.i.d. with copula C_θ
- [Chen & Fan \(2006\)](#): CML on the ranks of the *estimated* residuals $\hat{\varepsilon}_{it}$ has the same limit distribution as CML on the ranks of the true ε_{it}
- Monte Carlo: 500 samples, $T = 2000$, Gaussian copula $\rho = 0.8$, Student $t(5)$ innovations, GARCH as fitted to JPM and BAC
 - ▶ sd of $\hat{\rho}$: 0.0081 (true innovations) and 0.0082 (GARCH residuals); mean sandwich se 0.0081, mean inverse-Hessian se 0.0063
- Goodness of fit on residuals: [Rémillard \(2017\)](#) shows that rank-based tests keep their limit and the parametric bootstrap applies

Appendix: Nonparametric Tail Dependence, Conditions

- Schmidt & Stadtmüller (2006): $\hat{\lambda}_L = \frac{n}{k} \hat{C}_n\left(\frac{k}{n}, \frac{k}{n}\right)$ is consistent and asymptotically Normal if $k \rightarrow \infty$, $k/n \rightarrow 0$, the tail copula exists and a second-order bias condition holds ($\sqrt{k} A(n/k) \rightarrow 0$)
- The i.i.d. bootstrap of residual pairs assumes a static innovation copula; the stationary block bootstrap allows serial dependence in the tail indicators $1\{\text{lower corner}\} - 1\{\text{upper corner}\}$
- Mean block length by the automatic rule of Politis & White (2004), applied to the tail indicator; $B = 999$ draws, ranks recomputed in each draw

Appendix: Dependence in High Dimension and over Time

- DCC for large N : the full likelihood is slow and biased
 - ▶ Composite likelihood over pairs of assets, [Pakel, Shephard, Sheppard & Engle \(2021\)](#); DCC-NL, nonlinear shrinkage of the target $\bar{Q}$ (the shrinkage of Chapter 4), [Engle, Ledoit & Wolf \(2019\)](#)
- Vine (pair-copula) constructions, [Aas, Czado, Frigessi & Bakken \(2009\)](#)
 - ▶ $d(d-1)/2$ bivariate copulas, some conditional, in nested trees; each pair has its own family and tails
- Hierarchical Archimedean copulas, [Okhrin, Okhrin & Schmid \(2013\)](#)
 - ▶ Groups nested by the strength of dependence: banks within a country, countries within a region
- Dynamic copulas
 - ▶ Score-driven parameters ([Creal, Koopman & Lucas \(2013\)](#)) extend to many assets through factor copulas; [Oh & Patton \(2018\)](#) measure time-varying systemic risk in credit default swap spreads; conditional copulas: [Patton \(2006\)](#)

Appendix: From Correlations to Systemic Risk

- Correlation networks are the first step towards the systemic-risk measures of Chapter 18
 - ▶ Connectedness from Granger causality: [Billio, Getmansky, Lo & Pelizzon \(2012\)](#)
 - ▶ Spillover indices from variance decompositions: [Diebold & Yilmaz \(2009\)](#)
 - ▶ Tail-event networks: the Financial Risk Meter applied to EU energy firms, [Toma, Dima, Grecu, Pele & Lessmann \(2026\)](#)
- Energy prices and EU stock markets: dependence across regions and industrial groups, [Grecu, Cramer, Pele & Lessmann \(2025\)](#)
- Correlation networks inherit all the problems of correlations: non-linearity, time variation, tails

Appendix: Case Study, Data and Estimation

- **Model** (Eq. 8–11 of [Engle & Campos-Martins \(2023\)](#)): for $i \neq j$, $\Psi_{ij,t} = s_i s_j v_t$, $v_t = \text{Var}_{t-1}(x_t)$, so Ψ has one factor and its first principal component gives starting values (Eq. 12–13)
 - ▶ Equal loadings $s_i = 1$: the maximum-likelihood estimate is $\hat{x}_t = \frac{1}{N} \sum_i e_{it}^2$ (Eq. 17)
- **Estimation** (Section 5.3): maximise $-\frac{1}{2} \sum_{i,t} [\ln g_{it} + e_{it}^2/g_{it}]$, $g_{it} = s_i x_t + 1 - s_i$, alternating x_t and s_i until convergence
 - ▶ After each step: $s^\top s = 1$, mean of x_t equal to 1 (Remark 2); test ξ (Eq. 27): standardised mean of $(e_{it}^2 - 1)(e_{jt}^2 - 1)$ over the $N(N - 1)/2$ pairs, $\mathcal{N}(0, 1)$ under H_0
- **Data** (Section 7.1, Table B.1): daily USD log returns of 47 country ETFs listed on the NYSE, 2 June 2000 – 1 March 2021; same closing time, so no asynchrony; unbalanced panel
 - ▶ Mean: AR(1) plus two factors, the ACWI and PC1 of the ETF returns; GARCH(1,1) variances; COVOL fitted to the 47 residual series and the two factors
 - ▶ Also: 23 US asset-class series (Section 9) and Monte Carlo studies of the estimator and of the test (Sections 5.4, 6.1)

Appendix: Case Study, Key Results

Day	$\hat{x}_t$	$\bar{r}_t$ (%)	Event (label of the paper)
9 March 2020	51.416	−9.650	COVID-19: Italy locked down; oil price war
24 June 2016	49.190	−7.270	Brexit
5 August 2019	39.457	−2.574	Trade war: China devalues its currency
28 April 2020	38.883	2.334	COVID-19: tech share sell-off
17 September 2001	30.077	−8.630	NYSE reopens after 9/11

$\bar{r}_t$: mean return across the country ETFs; top five of the twenty largest values in Table 6 of [Engle & Campos-Martins \(2023\)](#)

- Loadings (Table 7): PC1 0.261 and ACWI 0.250 lead; France 0.217, Spain 0.200 against New Zealand 0.065, Poland 0.080: factors load more than idiosyncrasies, so correlations rise with COVOL (Eq. 41)
- ACWI squared residual on COVOL_m^2 (monthly mean of $\hat{x}_t$ minus 1): 0.326 (SE 0.025), $R^2 = 0.404$ (Table 8); mean pairwise ETF correlation +0.053 per unit (SE 0.009, Table 10, column 4); change in VIX 4.034 (SE 0.249), $R^2 = 0.517$ (Table 12)

Appendix: Case Study, Replication Protocol and Limitations

- **Replicate** the asset-class version (Section 9) with the US-listed ETFs of data/market, which all close at the same time: 11 sector ETFs, Russell 1000 growth and value, Russell 2000, developed and emerging markets outside the US, gold, long Treasuries, corporate bonds, crude oil
 - ▶ One factor, PC1; unbalanced panel; $\hat{x}_t$ by Eq. 17, then the iteration of Section 5.3; ξ before and after dividing e_{it}^2 by $\hat{g}_{it}$ (Section 7.2)
 - ▶ Benchmark: Table 15 ranks 9 November 2016 (43.04) and 9 March 2020 (40.47) among the top four; the dollar and the Russell 2000 growth/value split are not in the course data
- **Limitations stated by the authors**
 - ▶ After dividing by $\hat{g}_{it}$ the squares are still correlated: possibly a second factor (Section 7.2); multifactor COVOL is open (Section 9)
 - ▶ With $s^\top s = 1$ the loadings shrink as N grows: $R_s^2 = 0.640$ for $N = 50$, $T = 1000$ against 0.924 for $N = 10$ (Table 2); the test loses power with unequal loadings, so $T \geq 1000$ is advised
- After the sample of the paper: 3 April 2025 (16.5, -3.67%), after the US tariff announcement, and 10 November 2022 (15.6, 4.56%), a lower-than-expected US inflation print
 - ▶ Open question (Seminar C3): do elections, tariffs and inflation news share one factor, and does COVOL_m^2 explain the monthly DCC correlations beyond the change in VIX?

Appendix: The Discovery Loop, Reusable Prompts (1/2)

- Tools are interchangeable: an AI assistant (Claude, ChatGPT, Gemini, GitHub Copilot) and literature engines (Semantic Scholar, Elicit, Google Scholar)
- **Literature:** *"List studies since 2015 on the drivers of the stock–bond correlation, with DOI, sample, measure of correlation, identification strategy and main estimate."*
 - ▶ Resolve every DOI on Crossref and check the estimate in the paper, not in the summary
- **Hypotheses:** *"Give competing explanations for the 2022 sign change (inflation level, monetary-policy regime, term-premium shock, a one-off tightening) and what each predicts for ρ before 2022 and out of sample."*
- **Data and code:** *"Write Python that joins SPY and TLT adjusted prices from data/market on common days, computes monthly realised correlations and merges US CPI inflation published before each month starts."*
 - ▶ Replicate first: -0.40 (2002–2021) and 0.11 (2022–2026)

Appendix: The Discovery Loop, Reusable Prompts (2/2)

- **Robustness:** *“Write a pre-registered plan: the pre-2022 subsample, core CPI instead of headline, weekly instead of daily returns, DCC instead of realised correlation, control for realised volatility, HAC lag 12.”*
 - ▶ Placebo: replace π by simulated random walks of the same persistence; a persistent regressor can produce large t-statistics by chance
- **Critique:** *“Act as a referee: is b identified by a single episode, and would the Forbes–Rigobon volatility bias explain it? Give the test that would expose each.”*
- **Report:** every inflation measure, window and frequency tried; subsample estimates next to the full sample; any specification chosen after seeing results is exploratory and reported as such
- Code of the mini-case: [Quantlets/Ch_06/ai_discovery_case.py](#); inflation: US consumer price index for all urban consumers, U.S. Bureau of Labor Statistics

References (1/5)

- ▣ Aas, K., Czado, C., Frigessi, A., Bakken, H. (2009). Pair-copula constructions of multiple dependence. *Insurance: Mathematics and Economics*, 44(2), 182–198.
- ▣ Aielli, G.P. (2013). Dynamic conditional correlation: on properties and estimation. *Journal of Business & Economic Statistics*, 31(3), 282–299.
- ▣ Ang, A., Chen, J. (2002). Asymmetric correlations of equity portfolios. *Journal of Financial Economics*, 63(3), 443–494.
- ▣ Baker, S.R., Bloom, N., Davis, S.J. (2016). Measuring economic policy uncertainty. *Quarterly Journal of Economics*, 131(4), 1593–1636.
- ▣ Billio, M., Getmansky, M., Lo, A.W., Pelizzon, L. (2012). Econometric measures of connectedness and systemic risk in the finance and insurance sectors. *Journal of Financial Economics*, 104(3), 535–559.
- ▣ Bollerslev, T. (1990). Modelling the coherence in short-run nominal exchange rates: a multivariate generalized ARCH model. *Review of Economics and Statistics*, 72(3), 498–505.
- ▣ Bollerslev, T., Engle, R.F., Wooldridge, J.M. (1988). A capital asset pricing model with time-varying covariances. *Journal of Political Economy*, 96(1), 116–131.
- ▣ Caldara, D., Iacoviello, M. (2022). Measuring geopolitical risk. *American Economic Review*, 112(4), 1194–1225.
- ▣ Campbell, J.Y., Pflueger, C., Viceira, L.M. (2020). Macroeconomic drivers of bond and equity risks. *Journal of Political Economy*, 128(8), 3148–3185.
- ▣ Chen, X., Fan, Y. (2006). Estimation and model selection of semiparametric copula-based multivariate dynamic models under copula misspecification. *Journal of Econometrics*, 135(1–2), 125–154.
- ▣ Clayton, D.G. (1978). A model for association in bivariate life tables and its application in epidemiological studies of familial tendency in chronic disease incidence. *Biometrika*, 65(1), 141–151.
- ▣ Corsetti, G., Pericoli, M., Sbraccia, M. (2005). ‘Some contagion, some interdependence’: more pitfalls in tests of financial contagion. *Journal of International Money and Finance*, 24(8), 1177–1199.

References (2/5)

- ▣ Creal, D., Koopman, S.J., Lucas, A. (2013). Generalized autoregressive score models with applications. *Journal of Applied Econometrics*, 28(5), 777–795.
- ▣ Demarta, S., McNeil, A.J. (2005). The t copula and related copulas. *International Statistical Review*, 73(1), 111–129.
- ▣ Diebold, F.X., Mariano, R.S. (1995). Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13(3), 253–263.
- ▣ Diebold, F.X., Yilmaz, K. (2009). Measuring financial asset return and volatility spillovers, with application to global equity markets. *Economic Journal*, 119(534), 158–171.
- ▣ Embrechts, P., McNeil, A.J., Straumann, D. (2002). Correlation and dependence in risk management: properties and pitfalls. In M.A.H. Dempster (ed.), *Risk Management: Value at Risk and Beyond*, 176–223. Cambridge University Press.
- ▣ Engle, R.F. (2002). Dynamic conditional correlation: a simple class of multivariate generalized autoregressive conditional heteroskedasticity models. *Journal of Business & Economic Statistics*, 20(3), 339–350.
- ▣ Engle, R.F., Campos-Martins, S. (2023). What are the events that shake our world? Measuring and hedging global COVOL. *Journal of Financial Economics*, 147(1), 221–242.
- ▣ Engle, R.F., Kroner, K.F. (1995). Multivariate simultaneous generalized ARCH. *Econometric Theory*, 11(1), 122–150.
- ▣ Engle, R.F., Ledoit, O., Wolf, M. (2019). Large dynamic covariance matrices. *Journal of Business & Economic Statistics*, 37(2), 363–375.
- ▣ Engle, R.F., Sheppard, K. (2001). Theoretical and empirical properties of dynamic conditional correlation multivariate GARCH. NBER Working Paper 8554.
- ▣ Forbes, K.J., Rigobon, R. (2002). No contagion, only interdependence: measuring stock market comovements. *Journal of Finance*, 57(5), 2223–2261.
- ▣ Frank, M.J. (1979). On the simultaneous associativity of $F(x, y)$ and $x + y - F(x, y)$. *Aequationes Mathematicae*, 19, 194–226.

References (3/5)

- ▣ Franke, J., Härdle, W.K., Hafner, C.M. (2019). *Statistics of Financial Markets: An Introduction*, 5th ed. Springer.
- ▣ Genest, C., Ghoudi, K., Rivest, L.-P. (1995). *A semiparametric estimation procedure of dependence parameters in multivariate families of distributions*. *Biometrika*, 82(3), 543–552.
- ▣ Genest, C., Rémillard, B., Beaudoin, D. (2009). *Goodness-of-fit tests for copulas: a review and a power study*. *Insurance: Mathematics and Economics*, 44(2), 199–213.
- ▣ Genest, C., Rivest, L.-P. (1993). *Statistical inference procedures for bivariate Archimedean copulas*. *Journal of the American Statistical Association*, 88(423), 1034–1043.
- ▣ Grecu, R.A., Cramer, A.A., Pele, D.T., Lessmann, S. (2025). *The link between energy prices and stock markets in European Union countries*. *The North American Journal of Economics and Finance*, 78, 102420.
- ▣ Gumbel, E.J. (1960). *Bivariate exponential distributions*. *Journal of the American Statistical Association*, 55(292), 698–707.
- ▣ Härdle, W.K., Simar, L. (2019). *Applied Multivariate Statistical Analysis*, 5th ed. Springer.
- ▣ Herskovic, B., Kelly, B., Lustig, H., Van Nieuwerburgh, S. (2016). *The common factor in idiosyncratic volatility: quantitative asset pricing implications*. *Journal of Financial Economics*, 119(2), 249–283.
- ▣ Joe, H. (2005). *Asymptotic efficiency of the two-stage estimation method for copula-based models*. *Journal of Multivariate Analysis*, 94(2), 401–419.
- ▣ J.P. Morgan/Reuters (1996). *RiskMetrics – Technical Document*, 4th ed. New York.
- ▣ Kendall, M.G. (1938). *A new measure of rank correlation*. *Biometrika*, 30(1–2), 81–93.
- ▣ Li, D.X. (2000). *On default correlation: a copula function approach*. *Journal of Fixed Income*, 9(4), 43–54.

References (4/5)

- ▣ Longin, F., Solnik, B. (2001). [Extreme correlation of international equity markets](#). *Journal of Finance*, 56(2), 649–676.
- ▣ Mantegna, R.N. (1999). [Hierarchical structure in financial markets](#). *European Physical Journal B*, 11(1), 193–197.
- ▣ Markowitz, H. (1952). [Portfolio selection](#). *Journal of Finance*, 7(1), 77–91.
- ▣ Molenaar, R., Sénéchal, E., Swinkels, L., Wang, Z. (2024). [Empirical evidence on the stock–bond correlation](#). *Financial Analysts Journal*, 80(3), 17–36.
- ▣ Nelsen, R.B. (2006). [An Introduction to Copulas](#), 2nd ed. Springer.
- ▣ Newey, W.K., McFadden, D. (1994). [Large sample estimation and hypothesis testing](#). In R.F. Engle, D.L. McFadden (eds.), *Handbook of Econometrics*, Vol. 4, 2111–2245. Elsevier.
- ▣ Newey, W.K., West, K.D. (1987). [A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix](#). *Econometrica*, 55(3), 703–708.
- ▣ Oh, D.H., Patton, A.J. (2018). [Time-varying systemic risk: evidence from a dynamic copula model of CDS spreads](#). *Journal of Business & Economic Statistics*, 36(2), 181–195.
- ▣ Okhrin, O., Okhrin, Y., Schmid, W. (2013). [On the structure and estimation of hierarchical Archimedean copulas](#). *Journal of Econometrics*, 173(2), 189–204.
- ▣ Pakel, C., Shephard, N., Sheppard, K., Engle, R.F. (2021). [Fitting vast dimensional time-varying covariance models](#). *Journal of Business & Economic Statistics*, 39(3), 652–668.
- ▣ Patton, A.J. (2006). [Modelling asymmetric exchange rate dependence](#). *International Economic Review*, 47(2), 527–556.
- ▣ Pele, D.T., Wesselhöfft, N., Härdle, W.K., Kolossiatis, M., Yatracos, Y.G. (2023). [Are cryptos becoming alternative assets?](#). *European Journal of Finance*, 29(10), 1064–1105.

References (5/5)

- ▣ Politis, D.N., White, H. (2004). Automatic block-length selection for the dependent bootstrap. *Econometric Reviews*, 23(1), 53–70.
- ▣ Rémillard, B. (2017). Goodness-of-fit tests for copulas of multivariate time series. *Econometrics*, 5(1), 13.
- ▣ Rigobon, R. (2003). Identification through heteroskedasticity. *Review of Economics and Statistics*, 85(4), 777–792.
- ▣ Schmidt, R., Stadtmüller, U. (2006). Non-parametric estimation of tail dependence. *Scandinavian Journal of Statistics*, 33(2), 307–335.
- ▣ Sklar, A. (1959). Fonctions de répartition à n dimensions et leurs marges. *Publications de l'Institut de Statistique de l'Université de Paris*, 8, 229–231.
- ▣ Spearman, C. (1904). The proof and measurement of association between two things. *American Journal of Psychology*, 15(1), 72–101.
- ▣ Toma, F.-M., Dima, D.M., Grecu, R., Pele, D.T., Lessmann, S. (2026). Energy Financial Risk Meter: systemic risk in the EU energy sector. SSRN Working Paper.
- ▣ Tse, Y.K. (2000). A test for constant correlations in a multivariate GARCH model. *Journal of Econometrics*, 98(1), 107–127.
- ▣ Wied, D., Krämer, W., Dehling, H. (2012). Testing for a change in correlation at an unknown point in time using an extended functional delta method. *Econometric Theory*, 28(3), 570–589.